

Space

An algorithmic stereo reverb with room, chamber, hall, plate and spring.



Overview

Space is a stereo reverb in the spirit of the 90s desktop reverb units. ROOM, CHAMBER and HALL use early reflections followed by a dense eight-line tail, each sized for its space. PLATE has no early reflections: it is dense from the start, and bright. SPRING is a separate spring-tank model with two springs, one for each side, complete with the 'boing' chirp.

Around the reverb you get the usual shaping - pre-delay, decay, size, damping and modulation - plus tone filters on the reverb only, a vintage 500 Hz cut, ducking so the reverb steps back while you play, and FREEZE to hold the tail for ever.

Reach for it for

- Rooms and chambers for vocals, drums and acoustic instruments
- Halls and long ambient washes for pads and keys
- Bright plates on snares and vocals
- Surf and dub spring sounds on guitars and percussion
- Duckered reverb that stays out of the way of the dry part
- Frozen pads and drones (FREEZE)

Controls

Reverb type and display

ROOM / CHAMBER / HALL	Early reflections plus the reverb tail, from the smallest space to the largest.
PLATE	No early reflections; dense from the start and bright, with the modulation halved.
SPRING	A two-spring tank, left and right, with the characteristic chirp. EARLY and DIFFUSION do nothing here.
FREEZE	Holds the tail for ever: no new sound goes in and the reverb stops decaying. Switch it off to let the tail fade.
Window	A picture of what is ringing - the room, the plate or the springs - with the type's name.
Decay display	The pre-delay gap, the early reflections and the tail fading to -60 dB, with the DECAY time and pre-delay printed above (and FROZEN while FREEZE is on).
Meters	Output level, left and right, with a floating peak. Click a meter to clear the peak.

Reverb

PRE-DELAY	The gap before the reverb starts, 0 - 250 ms. Separates the reverb from the dry sound.
DECAY	How long the tail takes to fade by 60 dB, 0.2 - 20 s.
SIZE	The size of the space, 50 - 150%. For SPRING it sets the length of the springs.
DAMPING	How much sooner the highs die away than the lows, 0 - 100%. Higher is darker. For SPRING it darkens the tank.
MOD	Slow chorusing of the tail, 0 - 100%, so it shimmers rather than rings. Not used by SPRING.
EARLY	The level of the early reflections, 0 - 100% (ROOM, CHAMBER and HALL only; it reads n/a for PLATE and

	SPRING).
DIFFUSION	How smeared the onset is, 0 - 100%. Low settings let you hear separate echoes at the start; high settings blur them into a smooth wash. Not used by SPRING.
DUCK	Dips the reverb while you play and lets it bloom in the gaps, 0 - 100%.
WIDTH	The stereo width of the reverb, 0 - 100%.

Each knob shows its value while you hover over it or turn it; PRE-DELAY and DECAY are also shown all the time on the display.

Tone

These shape the reverb only - the dry sound is untouched.

LO CUT	Removes low end from the reverb, 20 - 1000 Hz. Keeps the tail from muddying the bass.
HI CUT	Removes top end from the reverb, 1 - 20 kHz.
0 / -3 / -6 / -9	The vintage reverb-send EQ: a cut of 3, 6 or 9 dB at 500 Hz, together with a high-pass at 140 Hz and a low-pass at 5 kHz. 0 switches it off.

Drift

DRIFT	Makes the tail's modulation wander slowly and randomly, 0 - 100% (not used by SPRING). 50% is the standard amount, 100% twice that, 0 none; the wander also gets a little faster as you turn it up.
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Master

MIX	Dry / reverb balance, 0 - 100% (30% by default). Up to 50% the dry sound stays at full level and the reverb comes in; above 50% the reverb stays at full level and the dry sound fades out. Use 100% on an FX send.
OUTPUT	Output level, -24 to +12 dB.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob (without dragging) switches GUARD on or off; its LED at the knob's upper right is dark when GUARD is off, red when it is on, and glows brighter while it is catching peaks.
GUARD	Off by default in this plugin. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

The header

PRESETS, arrows and display	PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. A preset sets the type and the reverb, tone (LO CUT, HI CUT) and DUCK controls and MIX, and switches FREEZE off; WIDTH, the vintage EQ, DRIFT, OUTPUT and the IN / OUT strip are left as they are. The last entry in the list resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Small Room	ROOM, 0.6 s decay, short pre-delay, plenty of early reflections.
Live Room	A bigger ROOM, 1.1 s, a little darker at the top.
Vocal Chamber	CHAMBER, 1.6 s with 15 ms pre-delay.
Concert Hall	HALL, 2.4 s with 25 ms pre-delay.
Cathedral (ducked)	A large HALL, 5.5 s with 40 ms pre-delay and DUCK at 30%, so it swells in the gaps.
Bright Plate	PLATE, 2 s, light damping and a 12 kHz HI CUT.
Snare Plate	A smaller PLATE, 1.2 s with no pre-delay.
Surf Spring	SPRING, 2 s, LO CUT 300 Hz, HI CUT 6 kHz.
Dub Spring	Longer, darker SPRING, 3 s with a 5 kHz HI CUT.
Ambient Wash	HALL at full size, 14 s decay, 60 ms pre-delay and heavy modulation, 40% mix.
Reset all to defaults	Every control back to its starting value.

Tips

- On an FX send, set MIX to 100% so only the reverb comes back.
- Raise LO CUT to 200 - 400 Hz on vocal and guitar reverbs to keep the low end clear.
- Try the -3 or -6 vintage EQ for the classic studio trick of a reverb that sits behind the mix without clouding it.
- DUCK lets you use a longer, bigger reverb on a busy part: it stays back while the part plays and blooms in the gaps.
- Turn FREEZE on at the end of a chord to hold it as a pad, then automate it off to let it fade.

File: gfx-space.jsfx